SUMMARY TABLE: CHEMISTRY GRADE 10

Sub-Strand 1.3: The Periodic Table — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 1.3: The Periodic Table
Driving Question	DRIVING QUESTION: Why do the 118 elements behave so differently, and how can knowing an element's position in the periodic table help us predict what it will do?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **The Mystery Element Card Challenge — Launching the Phenomenon**	Teacher distributes sets of 20 element cards (H to Ca) printed with atomic number, atomic mass, and key properties. Students attempt to arrange them into a logical grid on their desks. As groups compare arrangements, they notice that certain cards naturally cluster — elements that react violently with water land in one column, shiny soft metals in another, and gases that barely react at all in yet another. DQB NOTE: After the card sort, students post their opening 'I wonder…' and 'I notice…' sticky notes on the class Driving Questions Board. Teacher should photograph each group's card arrangement at this stage — this is the Lesson 1 baseline model for comparison at the end of the unit.	The 20 elements from Hydrogen to Calcium can be arranged into a two-dimensional grid of GROUPS (vertical columns) and PERIODS (horizontal rows). Elements in the same group share strikingly similar physical and chemical properties — for example, sodium (Na) and potassium (K) both react vigorously with water, while fluorine (F) and chlorine (Cl) are both reactive gases that form salts. The pattern is not accidental: position in the grid encodes information about how an element will behave. This is the foundation of the modern periodic table.	The card challenge is the anchoring phenomenon for the entire unit. Students should record: 'The element cards could be sorted because elements show a PERIODIC pattern — properties repeat at regular intervals as atomic number increases.' This directly begins to answer the driving question: elements behave differently because of differences in their internal structure, and the table's grid makes those differences visible. Pedagogical note — many students initially sort by atomic mass rather than atomic number; use this as a teachable moment foreshadowing Mendeleev's original challenge (Lesson 2). Cold-call at least 3 groups to share their sorting logic before revealing the accepted arrangement.
Lesson 2 **History of the Periodic Table: Standing on the Shoulders of Giants**	Students examine a timeline of key scientists — Döbereiner's triads (1817), Newlands' Law of Octaves (1865), Mendeleev's periodic law (1869), and Moseley's atomic number revision (1913). They analyse a facsimile of Mendeleev's original 1869 table, noting that he left	The periodic table was not invented in a single moment — it evolved over nearly 100 years as scientists proposed, tested, and revised models. Key milestones: Döbereiner noticed triads of elements with related properties; Newlands identified an 'octave' pattern; Mendeleev produced the	History shows that the periodic table is a scientific model — it was built on evidence, revised when evidence demanded it, and is still extended today (elements 113–118 were named between 2016–2022). This answers part of the driving question: elements were 'unknown' until discovered

	deliberate gaps and predicted properties of undiscovered elements (e.g., eka-silicon, later named Germanium). A short reading on Kenyan chemist contributions to modern materials science can be used to localise the history. Students compare Mendeleev's predictions for Germanium with the actual values when it was discovered in 1886.	first widely accepted table, arranged by atomic mass, and crucially left gaps for undiscovered elements; Moseley later corrected the arrangement by atomic number, resolving anomalies. Science is a self-correcting process — the table improved as new evidence emerged. Mendeleev's predictive gaps were confirmed when Gallium (1875) and Germanium (1886) were discovered with properties matching his forecasts.	because the table predicted blank spaces. Teacher note — draw an explicit parallel: just as Mendeleev predicted eka-silicon's density (~5.35 g/cm³) and the real Germanium measured 5.32 g/cm³, students can use the table TODAY to predict properties of unfamiliar elements. This is the power the driving question is pointing toward. Encourage students to update their DQB: 'I used to think the table was just a list… now I think it is a predictive tool.'
Lesson 3 **Electron Arrangement and Periodic Table Position**	Students use shell diagrams to draw electron arrangements for elements H to Ca (atomic numbers 1–20). They complete a structured table recording: element name, symbol, atomic number, number of shells occupied, and number of electrons in the outermost shell. They then colour-code their element cards from Lesson 1 by number of outer-shell electrons and observe that all cards in the same column share the same outer-shell electron count. Teacher directs: 'Count the columns from left — what do you notice about Group 1 elements? Group 2? Group 17?' Allow 2 minutes of wait time for independent noticing before cold-calling five students.	The group number equals the number of outer-shell (valence) electrons, and the period number equals the number of occupied electron shells. This is the invisible key that unlocks the entire periodic table. For example: Sodium (Na, atomic number 11) has electron arrangement 2,8,1 — 3 shells occupied (Period 3), 1 valence electron (Group 1). Chlorine (Cl, atomic number 17) has arrangement 2,8,7 — 3 shells (Period 3), 7 valence electrons (Group 17). Elements in the same group have the same number of valence electrons, which is why they behave similarly in chemical reactions.	Electron arrangement is the structural reason behind the periodic table's pattern — this is the core mechanistic explanation for the driving question. Elements behave differently because they have different numbers of valence electrons, which determines how they bond and react. Elements in the same column share valence electron count, hence similar behaviour. MODEL REVISION NOTE: Students should now revise their Lesson 1 card-sort model by annotating each element card with its electron arrangement (e.g., Na: 2,8,1). The model moves from 'sorted by observation' to 'sorted by internal structure.' Teacher should circulate and check that students correctly distinguish between total electrons and valence electrons — this is a high-frequency misconception.
Lesson 4 **Chemical Families: Identifying Groups and Trends in the Periodic Table**	Students watch a teacher demonstration (or video) of: (a) a small piece of sodium dropped into water in a trough — vigorous fizzing, the metal skates across the surface; (b) magnesium ribbon burned in air — brilliant white flame; (c) chlorine gas (generated safely) decolourising a damp litmus paper. Students record observations using the 'Claim–Evidence–Reasoning' framework. Teacher prompts: 'Where do sodium, magnesium, and chlorine sit on the periodic table? What pattern do you see between their position and their reactivity?' Students also examine samples or images of noble gas discharge tubes (neon signs — common in Nairobi's CBD — provide a vivid	The five major chemical families and their periodic table positions: (1) Alkali Metals — Group 1, very reactive, react violently with water to produce hydrogen gas and metal hydroxides (e.g., $2Na + 2H_2O \rightarrow 2NaOH + H_2$); (2) Alkaline Earth Metals — Group 2, reactive but less so than Group 1 (e.g., magnesium burns brightly in air); (3) Halogens — Group 17, very reactive non-metals, exist as diatomic molecules (F_2, Cl_2, Br_2, I_2), form salts with metals; (4) Noble Gases — Group 18, extremely unreactive, full outer shells (octet complete); (5) Transition Metals — Groups 3–12, variable oxidation states, often form coloured compounds, good conductors — iron (Fe)	Chemical families exist because elements in the same group have the same number of valence electrons — the same 'chemical personality.' This directly answers the driving question: an element's position tells us its family, its family tells us its reactivity, and reactivity determines what the element 'will do.' Teacher note — the sodium-water demonstration is high-impact; if live demonstration is not possible, use a vetted video and pause at key moments. Ask students: 'Why does potassium react even more violently than sodium with water?' (Answer: K has one more electron shell, so its valence electron is farther from the nucleus and more easily lost.) Students

	real-world context).	used in Kenyan construction, copper (Cu) in electrical wiring. Reactivity trends within groups: alkali metals become MORE reactive down the group (Li < Na < K); halogens become LESS reactive down the group (F > Cl > Br > I).	should add 'chemical family labels' to their evolving card-sort models.
Lesson 5 **Periodic Trends: Atomic Radius, Ionisation Energy, and Electronegativity**	Students analyse three data tables showing atomic radius (pm), first ionisation energy (kJ/mol), and electronegativity (Pauling scale) for Periods 2 and 3 elements and Group 1 and Group 17 elements. Working in pairs, they plot bar charts or line graphs of each property against atomic number and identify the direction of change across a period and down a group. Teacher asks: 'As you move from lithium to neon across Period 2, what happens to atomic radius? What happens to ionisation energy? Are these trends in the same direction or opposite?' Cold-call four students before providing the answer.	Three key periodic trends: (1) ATOMIC RADIUS — increases DOWN a group (more electron shells added, atom gets bigger) and DECREASES ACROSS a period left to right (more protons pull electrons closer to nucleus); (2) IONISATION ENERGY (energy needed to remove one valence electron) — decreases DOWN a group (valence electron farther from nucleus, easier to remove) and INCREASES ACROSS a period left to right (more protons hold electrons more tightly); (3) ELECTRONEGATIVITY (ability to attract bonding electrons) — decreases DOWN a group and INCREASES ACROSS a period left to right. Fluorine (F) is the most electronegative element in the entire periodic table. These three trends are interconnected and all trace back to two factors: number of proton charges in the nucleus and number of electron shells acting as 'shields.'	Periodic trends are the quantitative expression of the qualitative pattern students saw in Lesson 4. They explain WHY alkali metals become more reactive going down (ionisation energy decreases — easier to lose the valence electron) and WHY halogens become less reactive going down (electronegativity decreases — harder to gain an electron). This deepens the answer to the driving question: position predicts not just family behaviour but measurable, quantitative properties. MODEL REVISION NOTE: Students should annotate their table model with arrows showing the direction of each trend across periods and down groups. Teacher should check that students understand the nuclear charge vs. shielding interplay — a common misconception is that 'bigger atoms are always more reactive,' which is only true for metals, not non-metals.
Lesson 6 **Oxidation Numbers and Valency — Reading the Periodic Table's Charge Language**	Students receive a set of formula cards for common Kenyan-relevant compounds: NaCl (table salt used in cooking ugali), MgO (used in antacids), Al_2O_3 (aluminium oxide — component of Kenyan bauxite ore), $CaCl_2$ (used in water treatment), Na_2O, and K_2O. Working in groups, they use the known formula to back-calculate the charge each element must carry to make the compound electrically neutral. They then map these charges onto the periodic table and look for a pattern by group. Teacher directs: 'Look at the charge on Na in NaCl and K in KCl — what do you notice? Now look at the charge on Cl in both — what do you notice?'	An element's group number in the periodic table predicts its oxidation number (valency): Group 1 → +1; Group 2 → +2; Group 13 → +3; Group 14 → ±4; Group 15 → −3; Group 16 → −2; Group 17 → −1; Group 18 → 0 (noble gases do not typically bond). Transition metals (Groups 3–12) are the exception — they exhibit variable oxidation states (e.g., iron can be Fe^{2+} or Fe^{3+}, which is why iron forms two different types of oxide: FeO and Fe_2O_3, familiar as rust on Kenyan iron roofing sheets). Valency is the combining power of an element and determines the ratio in which elements combine to form compounds. The 'cross-multiply' rule: the valency of one element becomes the subscript of the other in the formula.	Oxidation numbers connect electron arrangement directly to chemical formula writing — if you know an element's group, you know its charge, and if you know the charges, you can write the formula. This is a powerful predictive tool that directly answers the driving question. Teacher note — students frequently confuse oxidation number with the number of valence electrons; clarify: valence electrons are the electrons available; oxidation number is the charge gained/lost/shared when bonding. Use the rust example (Fe_2O_3 vs. FeO) to illustrate why transition metals are 'wild cards.' Remind students that iron roofing (mabati) rusting in the Kenyan rain is a real-life example of Fe^{3+} forming Fe_2O_3. Students should update their models to

			include oxidation numbers next to each element symbol.
Lesson 7 **Metals, Non-Metals, and Metalloids — Reading the Periodic Table's Physical Map**	Students examine a set of physical property data cards for 12 elements (including iron, copper, sulphur, carbon, silicon, bromine, and argon) covering: lustre, electrical conductivity, thermal conductivity, malleability, melting point, and physical state at room temperature. They sort elements into metals, non-metals, and metalloids using the data, then compare their sort to the staircase line on the periodic table. Teacher prompts: 'Which elements sit right on or near the staircase line? What is unusual about their properties?' Students also examine silicon's role in solar panels — increasingly relevant to Kenya's growing solar energy sector (e.g., Garissa Solar Power Plant).	The periodic table is physically divided into metals (left and centre — roughly 80% of all elements), non-metals (upper right), and metalloids/semimetals (along the staircase boundary: B, Si, Ge, As, Sb, Te, Po). Metals are lustrous, malleable, ductile, good conductors of heat and electricity, and typically solid at room temperature (except mercury, Hg). Non-metals are generally poor conductors, brittle when solid, and can be gases, liquids, or solids at room temperature. Metalloids have intermediate properties — silicon (Si) is a semiconductor, making it essential for computer chips and solar panels. Hydrogen is a unique exception: it is a non-metal that appears in Group 1 because it has one valence electron, but it does not behave like an alkali metal.	The metal/non-metal/metalloid classification is a physical-property map that complements the chemical-family map from Lesson 4. Together, they give a complete picture of why elements 'behave so differently' — the first half of the driving question. An element's position (left, right, or on the boundary of the table) immediately signals its physical behaviour. Teacher note — use the Garissa Solar Power Plant as a motivating example: silicon's semiconducting property (metalloid character) makes photovoltaic cells possible. Ask students: 'If silicon were a true metal, would solar panels work the same way?' This bridges chemistry to technology and career pathways. MODEL REVISION NOTE: Students should shade their table model with three colours — one for metals, one for non-metals, one for metalloids.
Lesson 8 **Isotopes, Relative Atomic Mass, and the Periodic Table**	Students are given mass spectrometry data (simplified) for three elements: chlorine (Cl-35: 75%, Cl-37: 25%), carbon (C-12: 98.9%, C-13: 1.1%), and potassium (K-39: 93.3%, K-40: 0.01%, K-41: 6.7%). Working in pairs, they calculate the relative atomic mass (Ar) for each using the weighted average formula: Ar = Σ(isotope mass × fractional abundance). They then look up the Ar values on the periodic table and verify their calculations. Teacher asks: 'Why is the Ar of chlorine 35.5 — not a whole number? What does this tell us about what the periodic table value actually represents?'	Isotopes are atoms of the same element with the same number of protons but different numbers of neutrons (and therefore different mass numbers). The relative atomic mass (Ar) printed on the periodic table is NOT the mass of a single atom — it is the weighted average mass of all naturally occurring isotopes of that element, relative to 1/12 the mass of carbon-12. This is why chlorine's Ar is 35.5 (a mixture of Cl-35 and Cl-37 in a 3:1 ratio) rather than a whole number. Isotopes of the same element have identical chemical properties (same electron arrangement, same group, same valency) but different physical properties (different mass, different density, different nuclear stability). Radioactive isotopes (radioisotopes) have unstable nuclei — carbon-14 is used in radiocarbon dating of ancient Kenyan archaeological sites such as those at Olorgesailie.	Isotopes explain why atomic masses on the periodic table are not whole numbers — a detail students noticed in Lesson 1 but could not yet explain. This resolves a lingering puzzle from the anchoring phenomenon. The key insight: chemical behaviour is determined by proton/electron count (which is fixed for an element), not by neutron count — so isotopes of the same element sit in the same box on the table and behave identically in reactions. Teacher note — the carbon-14/Olorgesailie connection is an excellent local example of applied nuclear chemistry. Emphasise that despite having different masses, Cl-35 and Cl-37 would both form NaCl with sodium, in the same 1:1 ratio, because their electron arrangements are identical. This reinforces the core idea that position = electron arrangement = chemical behaviour.

Lesson 9 **Predicting Properties and Writing Chemical Formulas Using the Periodic Table**	Students work through a structured prediction challenge: they are given five 'unknown' elements identified only by their position on the periodic table (e.g., 'Element X is in Period 3, Group 16'). For each, they must predict: (a) number of valence electrons; (b) electron arrangement; (c) metal, non-metal, or metalloid; (d) oxidation number; (e) formula of its compound with sodium; (f) formula of its oxide. After completing predictions, students reveal the actual element identities and check their predictions against known data. Teacher uses the analogy of a Kenyan farmer predicting a crop's needs from the soil type — if you know the 'position' (soil type), you can predict the inputs needed.	The periodic table is a complete predictive system: knowing only an element's group and period number, a chemist can predict its electron arrangement, physical character (metal/non-metal), reactivity, oxidation number, and the formulas of compounds it will form. For example, an element in Period 3, Group 16 must be sulphur (S): electron arrangement 2,8,6; non-metal; oxidation number −2; formula with sodium → Na_2S; oxide formula → SO_2 or SO_3. This predictive power is why the periodic table is described as 'chemistry's most powerful organising tool.' Predictions integrate ALL prior lessons: electron arrangement (L3) → group/period position (L1) → chemical family (L4) → trends (L5) → oxidation number (L6) → formula writing (L6).	This lesson is the synthesis lesson — it demonstrates that the driving question has been fully answered in practice, not just in theory. Students can NOW use position to predict behaviour, which is exactly what the driving question asked. Teacher note — this lesson should feel like a 'reveal' or 'unlock' moment. Celebrate correct predictions enthusiastically. For students who struggle, prompt with the question sequence: 'What period? → How many shells? What group? → How many valence electrons? Metal or non-metal side? → What charge?' This structured questioning scaffold mirrors the NGSS practice of using models to predict phenomena. MODEL REVISION NOTE: Students should write a one-sentence 'position → prediction' rule on their model: 'If I know where an element sits, I know what it will do.'
Lesson 10 **Model Gallery Walk, DQB Closure & Unit Reflection**	Students display their final annotated periodic table models around the classroom for a Gallery Walk. Each model should show: electron arrangements, group/period labels, chemical family colour-coding, metal/non-metal/metalloid shading, periodic trend arrows, and oxidation numbers. As they walk, students use sticky dots to mark the model feature they find most insightful on each peer's display. The class then gathers at the Driving Questions Board: teacher reads each original 'I wonder…' question posted in Lesson 1 and asks, 'Can we answer this now?' Students tick off resolved questions and discuss any that remain open. The lesson closes with each student writing a final answer to the driving question — at least 5 sentences — in their exercise books.	Every element's behaviour is determined by its electron arrangement, which is encoded in its position on the periodic table (period = number of shells; group = number of valence electrons). This single rule explains: (a) why elements in the same group behave similarly (same valence electrons); (b) why reactivity trends exist across periods and down groups (nuclear charge vs. shielding); (c) why oxidation numbers are predictable (valence electrons lost, gained, or shared); (d) why the physical character of an element (metal/non-metal/metalloid) follows a spatial pattern on the table. The 118 elements are not 118 separate mysteries — they are a structured, predictable system whose logic is captured in one elegant grid. The periodic table is chemistry's most powerful predictive model.	The Gallery Walk and DQB closure ritualise the completion of the inquiry cycle: students began with wonder (L1 card sort), gathered evidence (L2–L9), and now publicly present explanations. This is the NGSS practice of 'Obtaining, Evaluating, and Communicating Information.' Teacher note — compare each student's Lesson 1 model (photograph taken in L1) with their Lesson 10 model on a split display if possible; the visual growth is a powerful motivator and assessment artefact. Final driving question answer should include: electron arrangement, group/period connection, at least one chemical family example, one periodic trend, and one real-world Kenyan application (e.g., iron in construction, sodium in cooking salt, silicon in solar panels, chlorine in water treatment). Award credit for models that show cumulative revision across all 10 lessons.